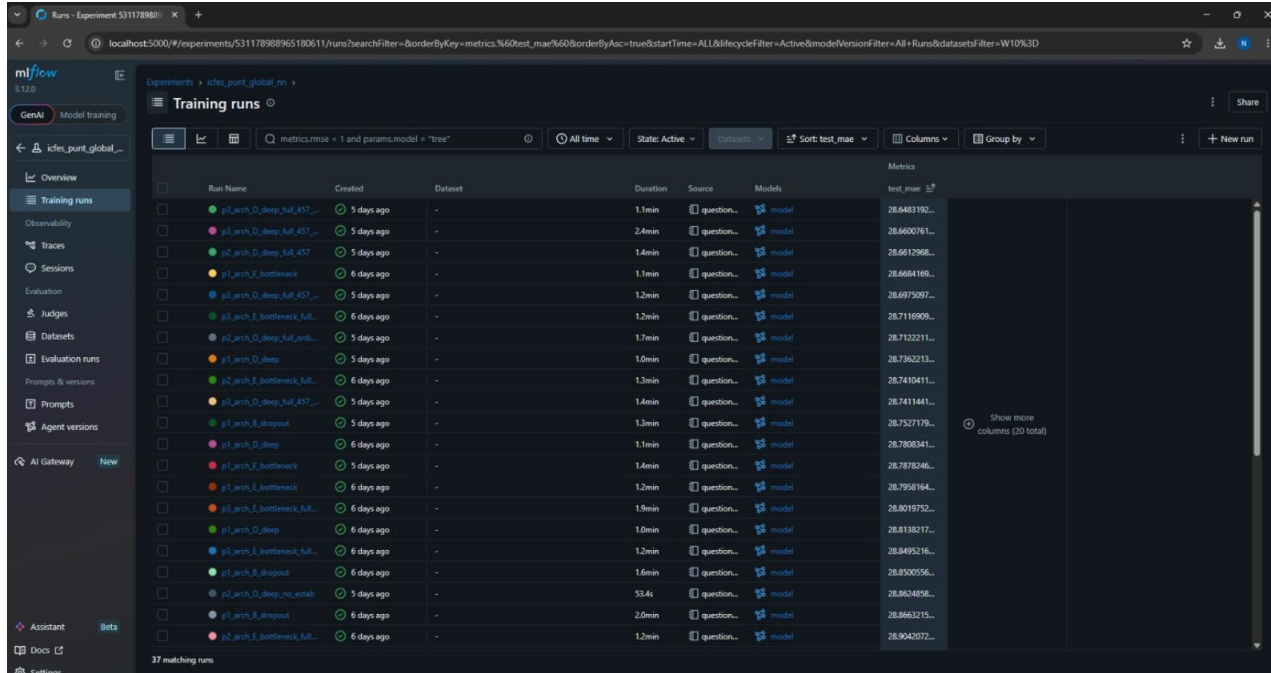


## SopORTE 2:

Los cuadernos de jupyter se encuentran en el repositorio de Git. A continuación, se presentan los soportes de las corridas de Mlflow para cada una de las preguntas de negocio propuestas:

### Primera pregunta de negocio:

Experimentos:



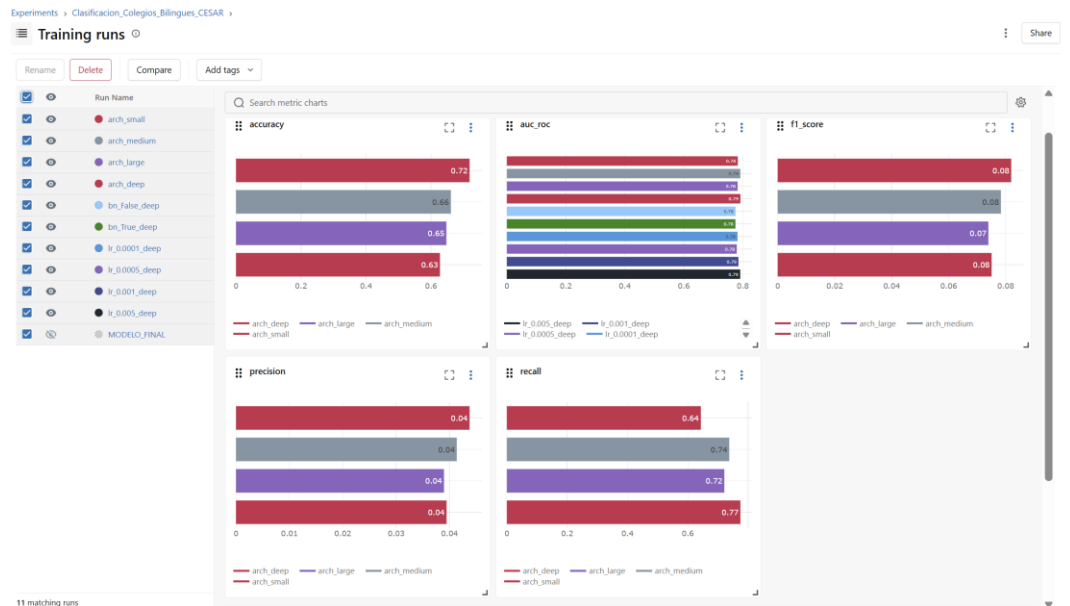
| Run Name                     | Created    | Dataset | Duration | Source      | Models | Metrics       |
|------------------------------|------------|---------|----------|-------------|--------|---------------|
| p3_arch_D_deep_full_457...   | 5 days ago | -       | 1.1min   | question... | model  | 28.6483192... |
| p3_arch_D_deep_full_457...   | 5 days ago | -       | 2.4min   | question... | model  | 28.6600761... |
| p3_arch_D_deep_full_457...   | 5 days ago | -       | 1.4min   | question... | model  | 28.6612966... |
| p1_arch_D_bottleneck...      | 6 days ago | -       | 1.1min   | question... | model  | 28.6684166... |
| p3_arch_D_deep_full_457...   | 5 days ago | -       | 1.2min   | question... | model  | 28.6975097... |
| p3_arch_D_bottleneck_full... | 6 days ago | -       | 1.2min   | question... | model  | 28.7116909... |
| p2_arch_D_deep_full_orin...  | 5 days ago | -       | 1.7min   | question... | model  | 28.7122211... |
| p1_arch_D_deep...            | 5 days ago | -       | 1.0min   | question... | model  | 28.7362213... |
| p3_arch_D_bottleneck_full... | 6 days ago | -       | 1.3min   | question... | model  | 28.7410411... |
| p1_arch_D_deep_full_457...   | 5 days ago | -       | 1.4min   | question... | model  | 28.7411441... |
| p1_arch_D_dropout...         | 5 days ago | -       | 1.3min   | question... | model  | 28.7527179... |
| p1_arch_D_deep...            | 6 days ago | -       | 1.1min   | question... | model  | 28.7803341... |
| p1_arch_D_bottleneck...      | 5 days ago | -       | 1.4min   | question... | model  | 28.7878246... |
| p1_arch_D_bottleneck...      | 6 days ago | -       | 1.2min   | question... | model  | 28.7958164... |
| p3_arch_D_bottleneck_full... | 6 days ago | -       | 1.9min   | question... | model  | 28.8019752... |
| p1_arch_D_deep...            | 6 days ago | -       | 1.0min   | question... | model  | 28.8138217... |
| p3_arch_D_bottleneck_full... | 6 days ago | -       | 1.2min   | question... | model  | 28.8495216... |
| p1_arch_D_dropout...         | 6 days ago | -       | 1.6min   | question... | model  | 28.8500556... |
| p2_arch_D_deep_no_actib...   | 5 days ago | -       | 51.4s    | question... | model  | 28.8624858... |
| p1_arch_D_dropout...         | 6 days ago | -       | 2.0min   | question... | model  | 28.8663215... |
| p3_arch_D_bottleneck_full... | 6 days ago | -       | 1.2min   | question... | model  | 28.9042072... |

Mae de entrenamiento y validación:



### Segunda pregunta de negocio:

Métricas para training runs



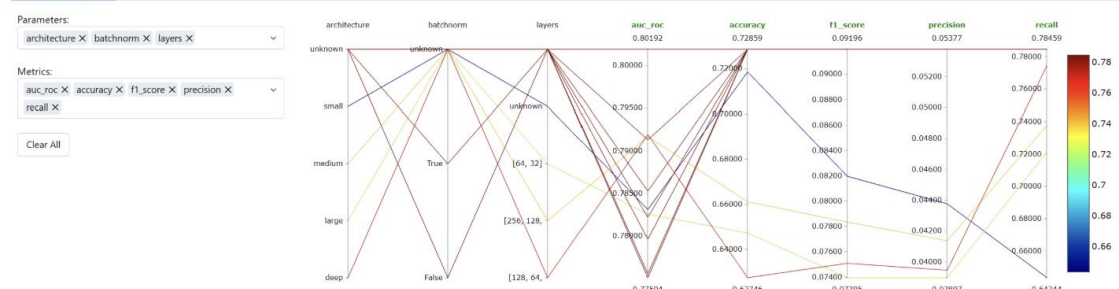
## Comparación de runs

Clasificacion\_Colegios\_Bilingues\_CESAR >

### Comparing 11 Runs from 1 Experiment

Visualizations

Parallel Coordinates Plot Scatter Plot Box Plot Contour Plot



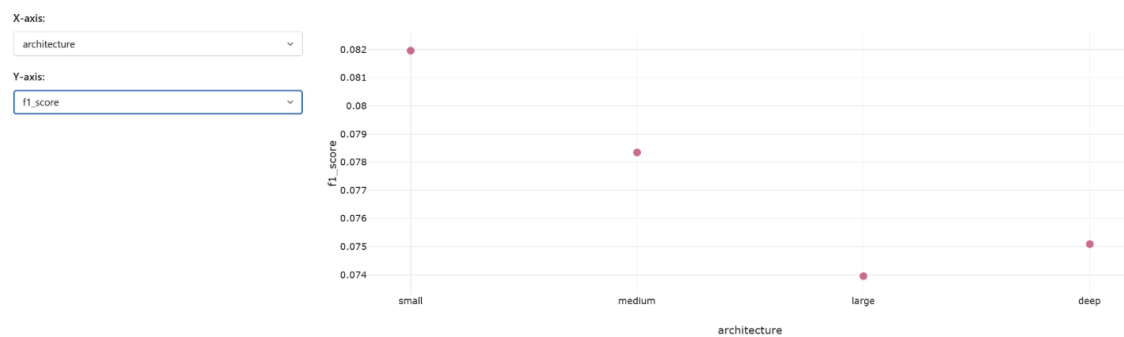
## Arquitecturas vs. f1-score

Clasificacion\_Colegios\_Bilingues\_CESAR >

### Comparing 11 Runs from 1 Experiment

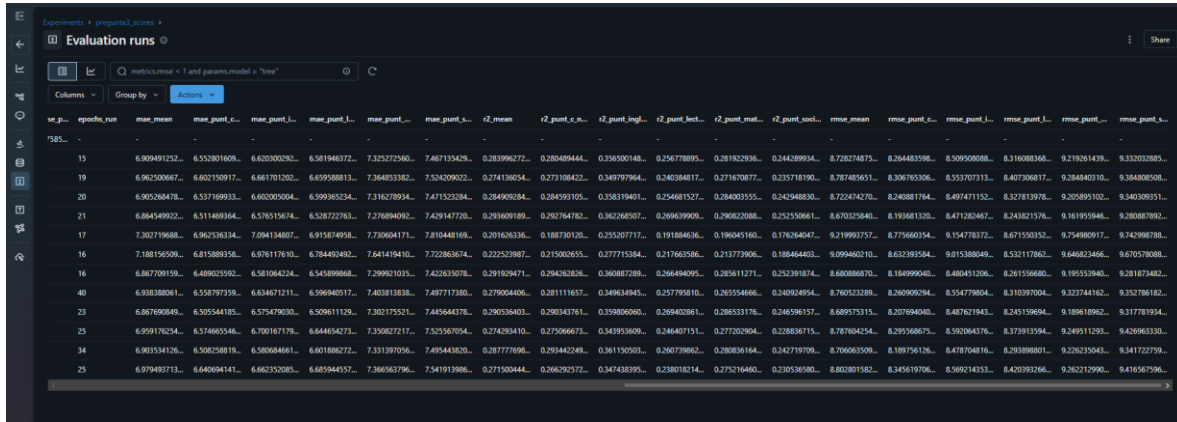
Visualizations

Parallel Coordinates Plot Scatter Plot Box Plot Contour Plot



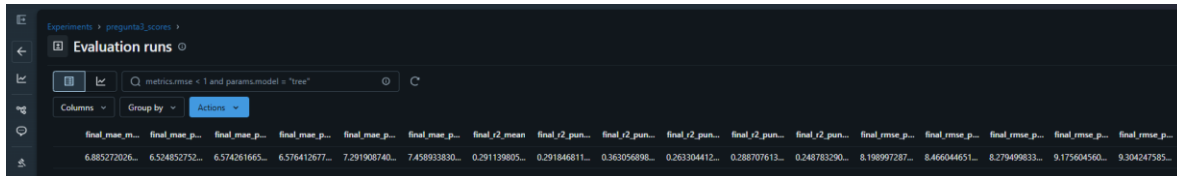
## Tercera pregunta de negocio:

Evaluation runs de arquitecturas diferentes al modelo final:



| run_id | epoch          | mean           | punt_c...      | punt_l...      | punt_l...      | punt_l...      | punt_s...      | r2_mean        | r2_punt_c...   | r2_punt_l...   | r2_punt_l...   | r2_punt_mat...  | r2_punt_soc... | r2_punt_soc... | r2_punt_soc... | r2_punt_soc... | r2_punt_soc... | r2_punt_soc... |
|--------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 15     | 6.909491252... | 6.552801609... | 6.820300292... | 6.581948372... | 7.325272560... | 7.467135429... | 0.283996772... | 0.280489444... | 0.356500148... | 0.256778895... | 0.281922936... | 0.244289934...  | 0.728274873... | 0.264483598... | 0.509508088... | 0.316088368... | 9.19261439...  | 9.33203385...  |
| 19     | 6.902500667... | 6.602150917... | 6.661701202... | 6.695988813... | 7.364853382... | 7.524209022... | 0.274136054... | 0.273108422... | 0.349797964... | 0.240384817... | 0.271670677... | 0.235771819...  | 0.787485651... | 0.306765306... | 0.553707313... | 0.407306817... | 9.284840310... | 9.364808508... |
| 20     | 6.90268478...  | 6.537169933... | 6.602000004... | 6.599365334... | 7.316278934... | 7.471525284... | 0.264906284... | 0.284593109... | 0.358319401... | 0.254681527... | 0.284003555... | 0.242948830...  | 0.722474270... | 0.240881764... | 0.497471152... | 0.327813978... | 9.202895102... | 9.340308351... |
| 21     | 6.864549922... | 6.511469364... | 6.576515674... | 6.528722763... | 7.276894092... | 7.429147720... | 0.293609189... | 0.262764782... | 0.362268507... | 0.269639909... | 0.290822088... | 0.252550661...  | 0.670325840... | 0.193681320... | 0.471282467... | 0.243821576... | 9.161959546... | 9.280887892... |
| 17     | 7.302719688... | 6.96236334...  | 7.094134807... | 6.915874958... | 7.730604171... | 7.810448169... | 0.201626336... | 0.188730126... | 0.255207717... | 0.191884636... | 0.196045160... | 0.176264047...  | 0.219993737... | 0.775660354... | 0.154778372... | 0.671550352... | 9.754980917... | 9.42996788...  |
| 16     | 7.188156509... | 6.815889358... | 6.976117610... | 6.784482490... | 7.641419410... | 7.722863874... | 0.222523987... | 0.215002655... | 0.277715384... | 0.217663586... | 0.213773906... | 0.188464403...  | 0.099480210... | 0.632393584... | 0.915388049... | 0.532117862... | 9.648823466... | 9.676578088... |
| 16     | 6.867709159... | 6.480255592... | 6.581064234... | 6.543898868... | 7.299921035... | 7.422635078... | 0.291820471... | 0.294263826... | 0.360887289... | 0.266494095... | 0.285611271... | 0.252391874...  | 0.680886670... | 0.184999040... | 0.480451206... | 0.261556660... | 9.195553940... | 9.281873482... |
| 40     | 6.938388061... | 6.56797319...  | 6.634671211... | 6.596940517... | 7.403813838... | 7.497717380... | 0.279004406... | 0.281111657... | 0.349634945... | 0.257795810... | 0.285554666... | 0.240924954...  | 0.760532389... | 0.268090894... | 0.354779804... | 0.310397004... | 9.327144162... | 9.352786182... |
| 23     | 6.86769049...  | 6.505346185... | 6.575479030... | 6.50961129...  | 7.302175521... | 7.445644378... | 0.290336403... | 0.290343761... | 0.359806060... | 0.269402861... | 0.286533176... | 0.246596157...  | 0.689575315... | 0.207694040... | 0.467621943... | 0.245159694... | 9.189618962... | 9.317781934... |
| 23     | 6.999178254... | 6.574665446... | 6.700167179... | 6.444654273... | 7.350821217... | 7.525367054... | 0.274293410... | 0.275066672... | 0.343953609... | 0.244607131... | 0.277202904... | 0.228838715...  | 0.787804354... | 0.295568675... | 0.592604376... | 0.373913594... | 9.429693339... | 9.429693339... |
| 34     | 6.93534126...  | 6.50825819...  | 6.580684661... | 6.601886272... | 7.331397056... | 7.495443820... | 0.287777896... | 0.293442249... | 0.361150503... | 0.267739662... | 0.280836164... | 0.2427719709... | 0.700663509... | 0.189756126... | 0.478704816... | 0.293898801... | 9.26235043...  | 9.341722759... |
| 25     | 6.979493713... | 6.640649141... | 6.662352085... | 6.685944557... | 7.366363796... | 7.541913988... | 0.271100444... | 0.266295572... | 0.347483395... | 0.238018214... | 0.275271640... | 0.230536580...  | 0.800201582... | 0.345619706... | 0.569214533... | 0.420393266... | 9.26221290...  | 9.416567596... |

Evaluation runs del modelo final:



| run_id         | final_mae_m... | final_mae_p... | final_mae_p... | final_mae_p... | final_mae_p... | final_mae_p... | final_r2_mean  | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... | final_r2_pun... |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 6.885272026... | 6.524852752... | 6.574261665... | 6.576412677... | 7.291908740... | 7.458933830... | 0.291139805... | 0.291846681... | 0.363056898...  | 0.263304412...  | 0.286707613...  | 0.246783290...  | 0.198997287...  | 0.466044651...  | 0.779499833...  | 0.175604560...  | 0.330427585...  | 9.304247585...  | 9.304247585...  |

Mae por prueba:

